

SUMTER, SC, USA

WMO: 720611

Lat: 34.000N

Lon: 80.367W

Elev: 55

StdP: 100.66

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00202

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-3.6	-13.9	1.1	-1.2	-11.2	1.4	1.9	8.5	15.9	7.7	13.6	1.1	350	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	37.2	27.5	35.1	25.7	33.8	25.2	32.0	34.2	28.7	31.7	26.4	32.1	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.3	29.5	33.9	27.6	23.7	30.8	25.1	20.3	28.4	110.7	34.2	93.0	34.9	82.6	32.0	35.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.3	5.4		-9.3	38.2	2.7	2.3	-11.3	39.9	-12.9	41.2	-14.4	42.5	-16.4	44.1	
			DB	-9.8	28.9	2.6	2.7	-11.7	30.8	-13.2	32.4	-14.6	33.8	-16.5	35.8	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	7.4	9.2	12.8	17.7	22.4	26.4	27.5	27.1	24.9	18.2	11.8	8.7
	DBStd	8.50	5.94	5.58	5.26	4.22	3.33	2.64	1.94	2.51	3.65	4.33	4.49	4.81
	HDD10.0	349	124	76	32	2	0	0	0	0	0	2	28	86
	HDD18.3	1416	341	261	183	62	9	0	0	0	1	56	202	301
	CDD10.0	3227	44	54	119	233	383	491	543	531	447	255	83	44
	CDD18.3	1253	2	5	12	43	134	241	285	272	198	52	7	1
	CDH23.3	12620	15	39	138	442	1248	2529	2952	2781	1972	464	36	4
	CDH26.7	5792	2	4	21	114	496	1234	1448	1386	961	124	2	0
Wind	WSAvg	1.9	2.0	2.3	2.5	2.5	2.1	1.8	1.7	1.4	1.6	1.7	1.7	1.9
Precipitation	PrecAvg	1199	107	94	108	79	89	140	137	128	97	74	68	80
	PrecMax	1720	210	193	297	230	206	364	350	292	247	391	221	185
	PrecMin	908	23	25	13	2	13	39	27	20	0	0	13	12
	PrecStd	206	51	46	59	49	42	72	66	62	56	71	43	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	27.3	28.8	31.3	35.1	39.1	37.5	39.9	40.9	32.1	27.0	23.0
		MCWB	19.4	18.8	18.2	21.3	23.9	27.2	25.4	31.7	32.7	23.8	20.6	17.4
	2%	DB	22.4	24.0	27.2	29.6	32.7	36.0	35.9	36.2	35.9	29.2	24.0	21.3
		MCWB	17.2	17.9	18.0	19.7	22.3	25.7	25.1	27.2	29.1	22.2	19.1	17.3
	5%	DB	19.9	22.1	24.9	27.8	31.3	33.9	34.1	34.0	33.0	27.7	22.5	19.1
		MCWB	16.1	16.2	16.8	19.6	22.1	24.8	25.3	26.0	26.0	21.0	18.4	15.9
	10%	DB	17.5	18.9	22.4	26.2	29.8	32.6	32.9	32.7	32.1	26.1	20.9	17.4
		MCWB	14.5	14.6	15.6	18.4	21.8	24.5	25.3	25.8	25.5	20.4	17.3	14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	20.8	20.3	23.6	25.8	28.7	27.4	33.7	33.7	25.5	23.0	20.0
		MCDB	23.1	24.2	25.0	28.2	30.6	36.2	32.7	37.4	37.5	28.5	23.4	22.1
	2%	WB	18.5	19.2	19.4	22.3	24.3	27.1	26.8	31.9	32.4	24.4	21.7	18.0
		MCDB	21.3	23.4	23.8	26.3	29.4	33.3	32.6	34.0	35.0	27.5	22.5	19.3
	5%	WB	17.0	17.5	18.5	21.3	23.5	26.0	26.2	27.9	30.0	23.3	19.2	16.9
		MCDB	19.1	19.7	23.1	25.0	28.4	32.0	32.1	31.5	31.9	26.0	22.0	18.7
	10%	WB	15.0	15.4	17.2	20.2	22.7	25.3	25.8	26.1	25.7	22.0	17.3	15.2
		MCDB	17.0	17.9	20.5	23.8	27.4	30.9	31.2	31.2	30.1	24.9	19.5	17.0
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.6	13.2	13.6	12.1	11.6	11.1	10.9	11.5	12.9	13.3	11.6
		MCDBR	14.6	15.0	15.8	14.8	14.5	13.0	12.3	12.2	13.3	14.0	14.3	14.0
		MCWBR	9.2	9.1	7.8	6.8	5.5	4.5	4.2	4.3	5.3	7.0	8.8	10.2
	5% WB	MCDBR	11.1	13.4	12.8	11.3	11.5	11.9	11.0	10.8	9.6	10.4	11.0	12.4
		MCWBR	8.5	8.9	7.1	6.2	5.0	4.5	4.3	4.6	4.8	5.8	7.7	9.9
Clear-Sky Solar Irradiance	taub	0.319	0.328	0.363	0.410	0.474	0.505	0.552	0.521	0.438	0.393	0.343	0.315	
	taud	2.500	2.472	2.400	2.259	2.130	2.086	1.977	2.054	2.274	2.369	2.469	2.549	
	Ebn at Noon	893	922	911	878	823	794	753	770	824	836	855	878	
	Edh at Noon	85	97	112	136	156	163	181	165	126	106	87	77	
All-Sky Solar Radiation	RadAvg	2.77	3.41	4.50	5.66	6.15	6.40	6.10	5.52	4.75	3.97	3.05	2.44	
	RadStd	0.15	0.36	0.34	0.40	0.48	0.44	0.38	0.27	0.41	0.48	0.24	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+186	+132

Nomenclature: See separate page